Lecture 6

Suggested Reading - Section (2.5-2.7) (Pai) - for today.
3.1-3.2 (next class).

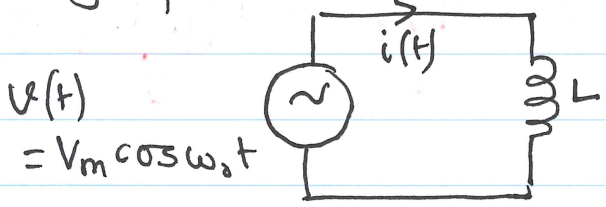
Last time : * Why three phase ?
* How to solve three-phase Y-connected systems?

Today : * Revist a "basic" concept.
* Δ-connected three-phase systems.

Q. So we have been using/"memorizing"
1. "in a R-L load, voltage leads current.
2. in a R-C load, current leads voltage"
3. Q. is (+)ve in L (inductor).
4. Q is (-)ve in C (capacitor).

Let us try to understand instead of memorizing. (build an intuition). I will show with inductor. Try yourself with a pure capacitor.

Imagine, a pure inductor connected to a single-phase ac source.



$$v(t) = L\frac{di}{dt}$$

$\Downarrow$ Fourier.

$$V(j\omega) = j\omega L\, I(j\omega)$$

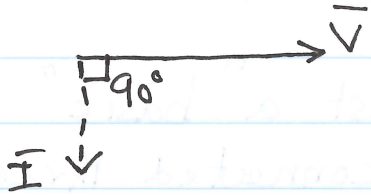
phasors ↙ ↙ phasor

$\bar{V} = (j\omega L)\,\bar{I} \quad \Rightarrow \bar{I} = \frac{1}{j\omega L}\,\bar{V}$

↑ 90° operator

$= -j\left(\frac{1}{\omega L}\right)\bar{V}$

↓ $-90°$ phase shift



current is lagging voltage by 90°.

[If it was pure R, current would have been in phase with $\bar{V}$. Therefore, any R-L, load something in between.]

Q. What do we expect with sign of reactive power?

Complex power, $\bar{S} = \bar{V}\bar{I}^*$

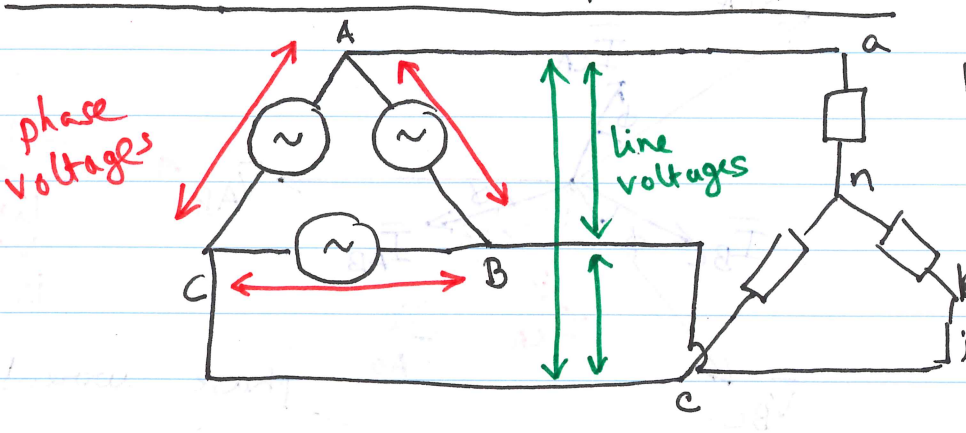
$= \bar{V}\left[-j\frac{1}{\omega L}\cdot\bar{V}\right]^*$

$= |\bar{V}|^2 \frac{1}{\omega L}\cdot[j]$

$P + jQ = j\frac{|\bar{V}|^2}{\omega L}$ positive sign of Q

obviously $P = 0$.

Delta-connected three phase systems : We can also take three single phase sources and form a Δ instead of Y.

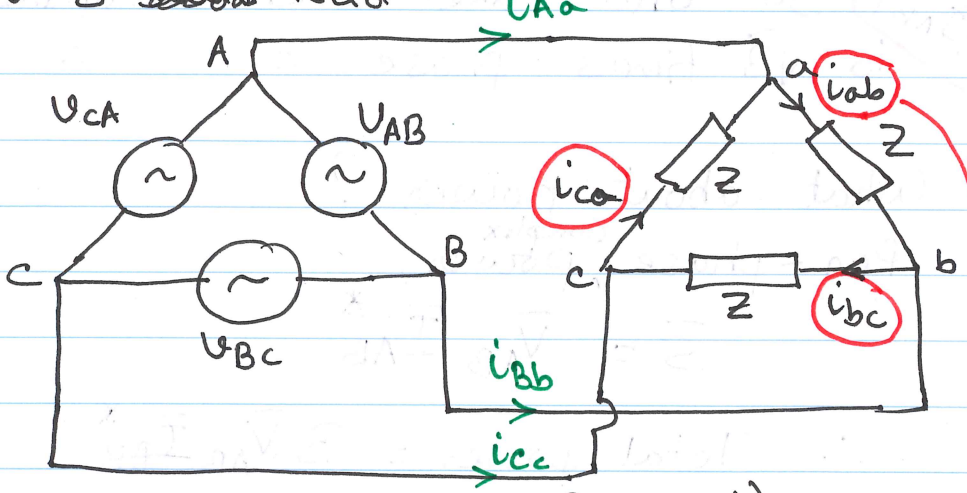


Let us call, v_{AB} our reference.

$$v_{AB}(t) = V_m \cos(\omega_0 t)$$
$$v_{BC}(t) = V_m \cos(\omega_0 t - 120^\circ)$$
$$v_{CA}(t) = V_m \cos(\omega_0 t + 120^\circ)$$

} Phase voltages. = Line voltages.

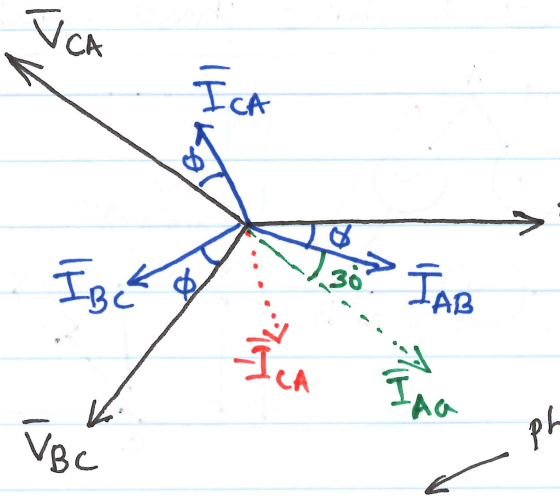
What happens if we connect a Δ-source to Δ load?



Note: there is no neutral.

Phase currents

$$i_{ab}(t) = I_m \cos(\omega_0 t - \phi)$$
$$i_{bc}(t) = I_m \cos(\omega_0 t - \phi - 120^\circ)$$
$$i_{ca}(t) = I_m \cos(\omega_0 t - \phi - 240^\circ)$$



Note our line voltage is reference.

phase currents.

So, what is, $\bar{I}_{Aa} = \bar{I}_{AB} - \bar{I}_{CA}$ [KCL at node a].

↑ line current

So, in a Δ-connected system, line current lags phase current by 30° and line current magnitude is $\sqrt{3}$ times phase "

$\angle \bar{I}_{Aa} = \angle \bar{I}_{AB} - 30^\circ$

$|\bar{I}_{Aa}| = \sqrt{3}|\bar{I}_{AB}|$

Q. What about power?

Per-phase complex power,

$$\bar{S} = \bar{V}_{AB}\bar{I}_{AB}^*$$

$\therefore$ Total power $= 3\bar{V}_{AB}\bar{I}_{AB}^*$

$= 3|\bar{V}_{AB}||\bar{I}_{AB}|(\cos\phi + j\sin\phi)$

where ϕ is the phase angle b/w $\bar{V}_{AB}$ and $\bar{I}_{AB}$